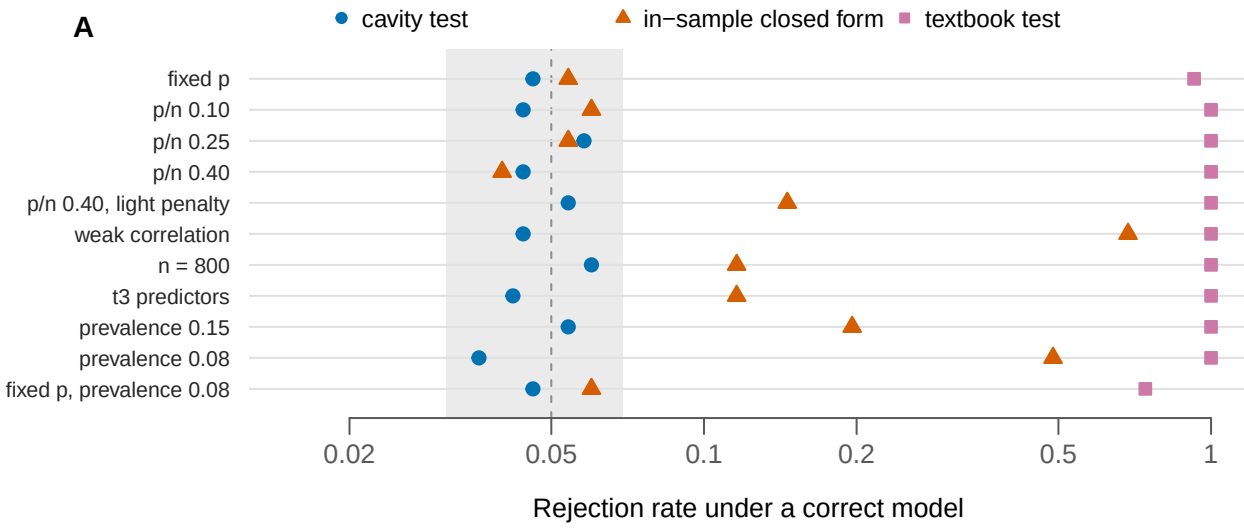
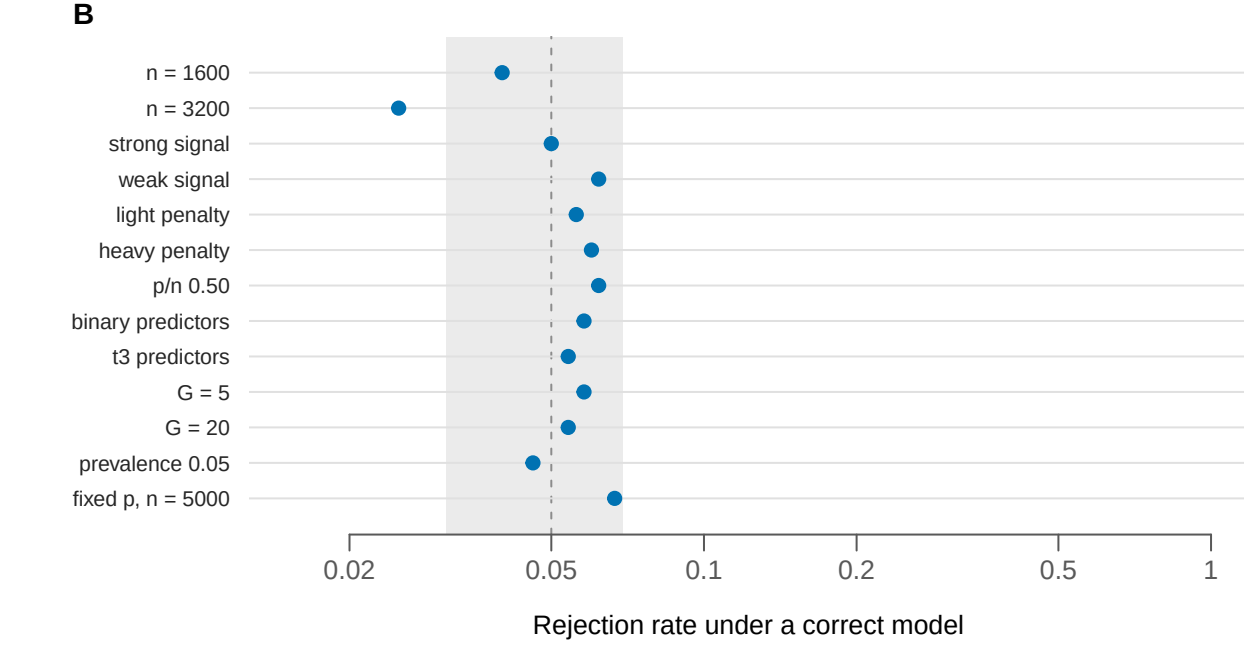
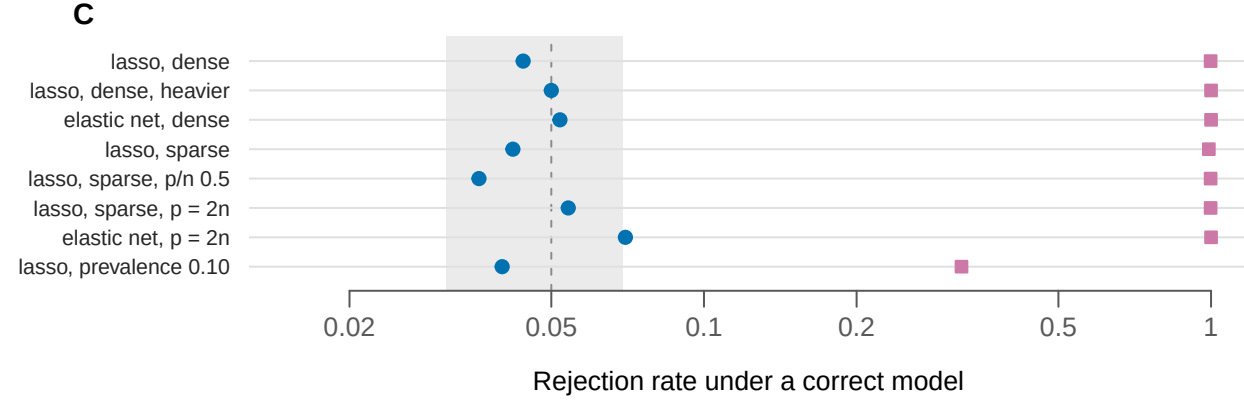

A
cavity test
in−sample closed form
textbook test
fixed p
p/n 0.10
p/n 0.25
p/n 0.40
p/n 0.40, light penalty
weak correlation
n = 800
t3 predictors
prevalence 0.15
prevalence 0.08
fixed p, prevalence 0.08
0.02
0.05
0.1
0.2
0.5
1
Rejection rate under a correct model


B
n = 1600
n = 3200
strong signal
weak signal
light penalty
heavy penalty
p/n 0.50
binary predictors
t3 predictors
G = 5
G = 20
prevalence 0.05
fixed p, n = 5000
0.02
0.05
0.1
0.2
0.5
1
Rejection rate under a correct model


C
lasso, dense
lasso, dense, heavier
elastic net, dense
lasso, sparse
lasso, sparse, p/n 0.5
lasso, sparse, p = 2n
elastic net, p = 2n
lasso, prevalence 0.10
0.02
0.05
0.1
0.2
0.5
1
Rejection rate under a correct model